

Lecture 7: Fourier Transform

CSE 219: Signals and Linear Systems

Anik Saha

Lecturer, Department of CSE, BUET

Fourier Transform: to ∞ and beyond

Looking inside any function. Yes, not just periodic ones!

Aperiodic functions crying in the background...



Fourier Transform

When the period goes to infinity!

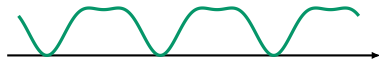
Fourier series decomposes **periodic** signals.

Fourier transform lets us do something similar for **aperiodic** signals.

Fourier series had one big condition

Fourier Series

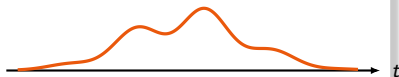
$$x(t + T) = x(t)$$



keeps repeating

But real signals
often do not

no fixed period



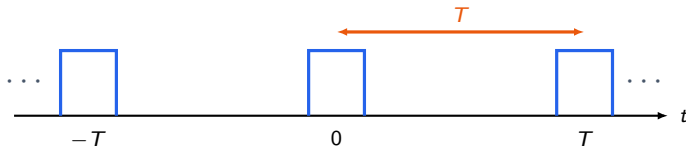
one-time event

Can we still describe a non-repeating signal using frequencies?

Let's cheat first: repeat one lonely pulse

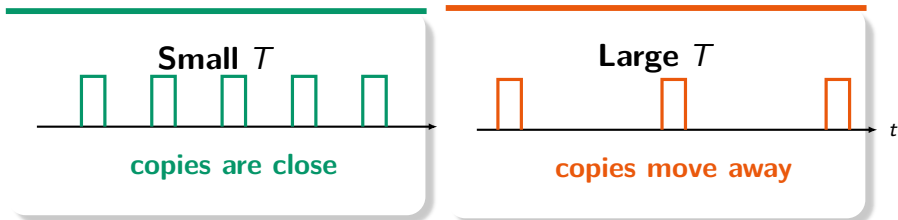
Start with one finite-duration signal $x(t)$

$$\tilde{x}_T(t) = \sum_{m=-\infty}^{\infty} x(t - mT)$$



Now it is periodic, so Fourier series works.

Now stretch the period

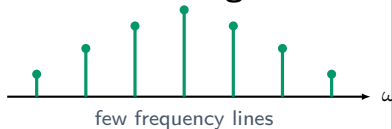


As $T \rightarrow \infty$, the repeated signal starts looking like one isolated signal.

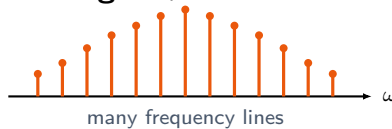
What happens in frequency?

$$\omega_0 = \frac{2\pi}{T}, \quad \Delta\omega = \omega_0.$$

Small T , large $\Delta\omega$



Large T , small $\Delta\omega$



Increasing T makes the allowed frequencies closer and closer.

Fourier-series coefficients of the repeated signal

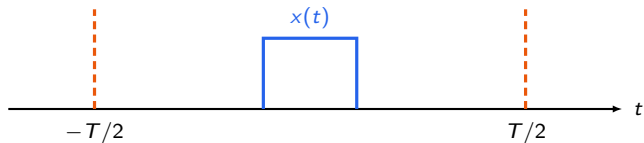
The periodic signal has Fourier series

$$\tilde{x}_T(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$a_k = \frac{1}{T} \int_{\text{one period}} \tilde{x}_T(t) e^{-jk\omega_0 t} dt$$

This is still the same Fourier-series coefficient formula.

Choose one period that contains the pulse



$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}_T(t) e^{-jk\omega_0 t} dt$$

Inside this period, $\tilde{x}_T(t)$ is just the isolated pulse $x(t)$.

$$\frac{1}{T} \int_{-\infty}^{\infty} x(t) e^{-jk\omega_0 t} dt$$

A smooth spectrum is hiding behind the stems

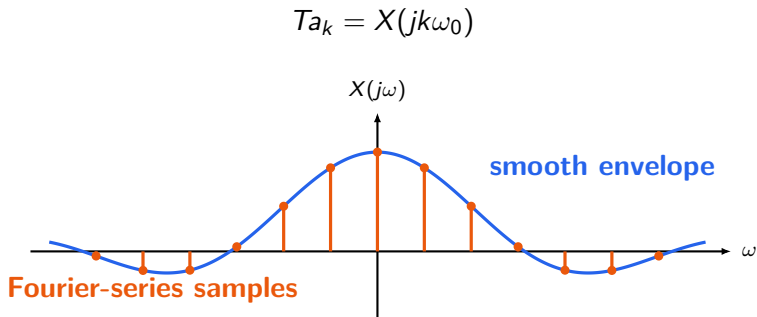
Look at the integral part

$$a_k = \frac{1}{T} \underbrace{\int_{-\infty}^{\infty} x(t) e^{-jk\omega_0 t} dt}_{\text{depends on } k\omega_0}$$

Let

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

So the stems are samples of a smooth curve



As T increases, the sample locations $k\omega_0$ become closer.

But why do raw coefficients shrink?

$$a_k = \frac{1}{T} X(jk\omega_0)$$

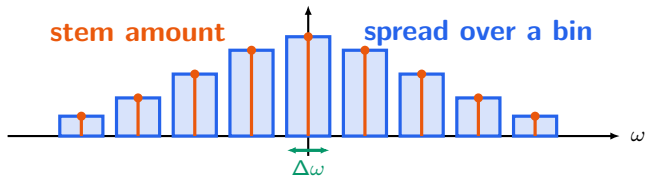
Since

$$\Delta\omega = \omega_0 = \frac{2\pi}{T},$$

we have

$$\frac{1}{T} = \frac{\Delta\omega}{2\pi}.$$

Think of it like frequency-density



When $\Delta\omega \rightarrow 0$, the bins become thinner and thinner, and the discrete spectrum becomes a continuous spectrum.

The Fourier-series sum becomes an integral

Start from Fourier-series synthesis

$$\tilde{x}_T(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

Use

$$a_k = \frac{\Delta\omega}{2\pi} X(jk\omega_0).$$

$$\tilde{x}_T(t) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(jk\omega_0) e^{jk\omega_0 t} \Delta\omega$$

Let $T \rightarrow \infty$

$$T \rightarrow \infty \quad \Rightarrow \quad \omega_0 = \Delta\omega = \frac{2\pi}{T} \rightarrow 0$$

$$k\omega_0$$

becomes a continuous variable

$$\omega$$

$$\Sigma$$

becomes

$$\int$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

The Fourier transform pair

Analysis

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) \longrightarrow X(j\omega)$$

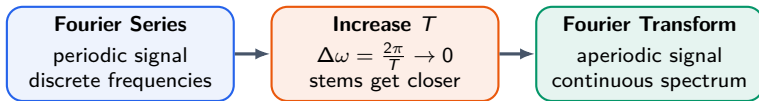
Synthesis

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

$$X(j\omega) \longrightarrow x(t)$$

Fourier transform is Fourier series after the frequency lines become continuous.

The whole story in one picture



Fourier transform is not a totally new idea.
It is the continuous-frequency version of Fourier series.

Let's keep rotating and oscillating random things...!

McDonald's



WcDonald's



EcDonald's



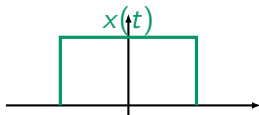
3cDonald's



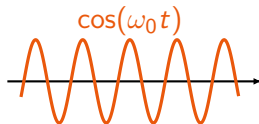
Injecting oscillation into a signal

$$x(t) \longrightarrow x(t) \cos(\omega_0 t)$$

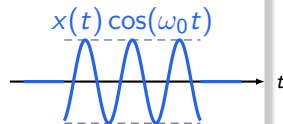
Envelope $x(t)$



Oscillation



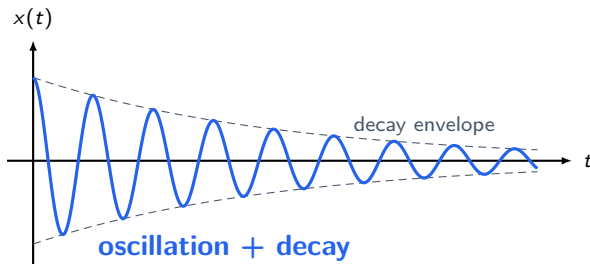
Oscillating envelope



We keep the shape of $x(t)$ as an envelope, but put oscillation inside it.

This happens in real systems too

A damped oscillation



Examples: damped pendulum, vibrating string with loss, ringing circuits, fading echoes.

So what transform should we expect?

$$x(t) \longleftrightarrow X(j\omega).$$

What is the Fourier transform of

$$x(t) \cos(\omega_0 t)?$$

Let's first solve the cleaner version:

$$x(t)e^{j\omega_0 t}.$$

Multiplying by $e^{j\omega_0 t}$ shifts every frequency

A frequency component inside $x(t)$ looks like

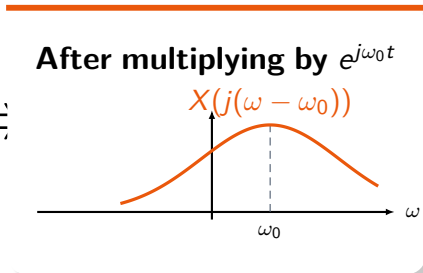
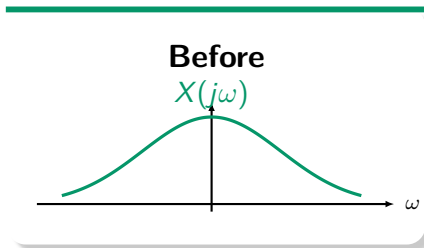
$$e^{j\omega t}.$$

After multiplying by $e^{j\omega_0 t}$,

$$e^{j\omega t} \cdot e^{j\omega_0 t} = e^{j(\omega + \omega_0)t}.$$

Every component at ω moves to $\omega + \omega_0$.

The spectrum slides to the right



To move the spectrum right by ω_0 , we write $X(j(\omega - \omega_0))$.

Frequency-shifting property

If

$$x(t) \longleftrightarrow X(j\omega),$$

then

$$x(t)e^{j\omega_0 t} \longleftrightarrow X(j(\omega - \omega_0))$$

Multiplication by a complex exponential shifts the spectrum.

Cosine is two rotating exponentials

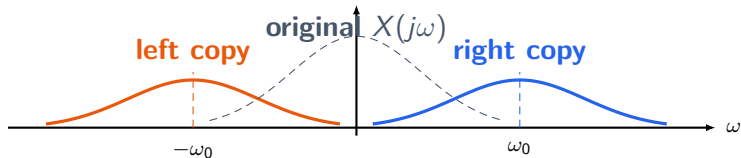
$$\cos(\omega_0 t) = \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t})$$

So

$$x(t) \cos(\omega_0 t) = \frac{1}{2} x(t) e^{j\omega_0 t} + \frac{1}{2} x(t) e^{-j\omega_0 t}.$$

A cosine creates two shifted copies of the spectrum.

Cosine modulation creates two spectral copies



$$x(t) \cos(\omega_0 t) \quad \longleftrightarrow \quad \frac{1}{2}X(j(\omega - \omega_0)) + \frac{1}{2}X(j(\omega + \omega_0))$$

What did oscillation do?

In time:

$$x(t) \rightarrow x(t) \cos(\omega_0 t)$$

We injected oscillation.

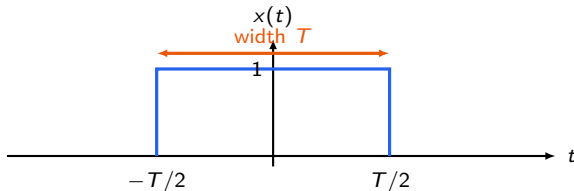
In frequency:

$X(j\omega) \rightarrow$ two shifted copies

The spectrum
moves to $\pm\omega_0$.

A very common transform: the rectangle

$$x(t) = \begin{cases} 1, & -\frac{T}{2} < t < \frac{T}{2}, \\ 0, & \text{otherwise.} \end{cases}$$



This transform appears everywhere, let's keep it in our toolbox :D

Fourier transform of the rectangle

Since $x(t) = 1$ only from $-T/2$ to $T/2$,

$$X(j\omega) = \int_{-T/2}^{T/2} e^{-j\omega t} dt.$$

$$X(j\omega) = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-T/2}^{T/2} = \frac{e^{j\omega T/2} - e^{-j\omega T/2}}{j\omega}.$$

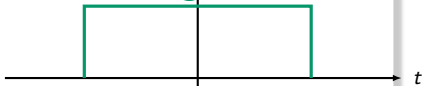
$$X(j\omega) = \frac{2 \sin(\omega T/2)}{\omega}$$

Wider rectangle, narrower spectrum

$$x(t) = 1, \quad |t| < \frac{T}{2} \quad \Longleftrightarrow \quad X(j\omega) = \frac{2 \sin(\omega T/2)}{\omega}.$$

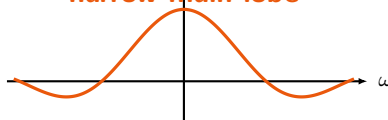
Time domain

large T



Frequency domain

narrow main lobe



Practice: two centered rectangles

$$x_1(t) = \begin{cases} 1, & -1 < t < 1, \\ 0, & \text{otherwise.} \end{cases}$$

$$X_1(j\omega) = ?$$

$$x_2(t) = \begin{cases} 1, & -3 < t < 3, \\ 0, & \text{otherwise.} \end{cases}$$

$$X_2(j\omega) = ?$$

Use

$$\text{width } T \implies X(j\omega) = \frac{2 \sin(\omega T/2)}{\omega}.$$

Answer: just plug in the width

For $-1 < t < 1$, the width is $T = 2$:

$$X_1(j\omega) = \frac{2 \sin(\omega)}{\omega}.$$

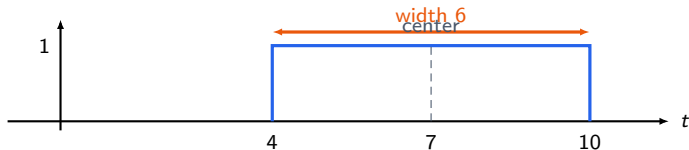
For $-3 < t < 3$, the width is $T = 6$:

$$X_2(j\omega) = \frac{2 \sin(3\omega)}{\omega}.$$

The same rectangle formula works once we identify the width.

What if the rectangle is shifted?

$$x(t) = \begin{cases} 1, & 4 < t < 10, \\ 0, & \text{otherwise.} \end{cases}$$



This is a width-6 centered rectangle shifted right by 7.

Answer: rectangle formula plus time shift

A width-6 rectangle centered at zero has transform

$$R(j\omega) = \frac{2 \sin(3\omega)}{\omega}.$$

Our signal is

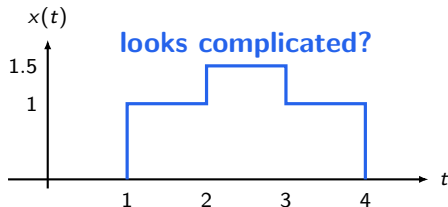
$$x(t) = r(t - 7).$$

Using time shifting,

$$r(t - t_0) \longleftrightarrow e^{-j\omega t_0} R(j\omega).$$

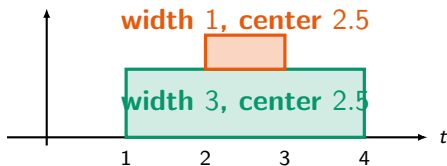
A Lego block signal

$$x(t) = \begin{cases} 1, & 1 < t < 2, \\ 1.5, & 2 < t < 3, \\ 1, & 3 < t < 4, \\ 0, & \text{otherwise.} \end{cases}$$



Decompose the Lego block

$$x(t) = \underbrace{\text{height 1 rectangle from 1 to 4}}_{\text{wide block}} + \underbrace{\text{height 0.5 rectangle from 2 to 3}}_{\text{small top block}}.$$



Fourier transform of the Lego block

Wide block:

$$\text{width 3, center 2.5} \implies e^{-j2.5\omega} \frac{2 \sin(1.5\omega)}{\omega}.$$

Top block:

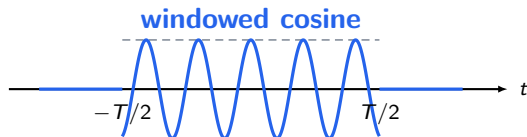
$$\text{height 0.5, width 1, center 2.5} \implies 0.5e^{-j2.5\omega} \frac{2 \sin(0.5\omega)}{\omega}.$$

$$X(j\omega) = e^{-j2.5\omega} \left[\frac{2 \sin(1.5\omega)}{\omega} + \frac{\sin(0.5\omega)}{\omega} \right]$$

Now lock a cosine inside the rectangle

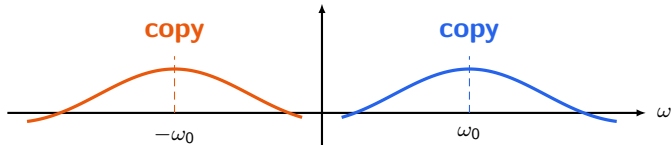
Let $r(t)$ be the rectangle from $-T/2$ to $T/2$.

$$x(t) = r(t) \cos(\omega_0 t)$$



This is just modulation of a rectangle.

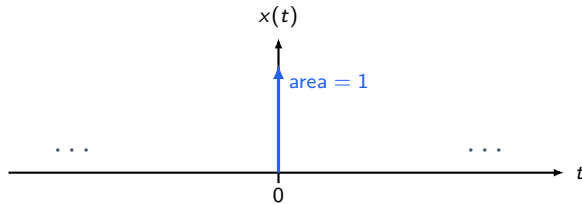
A windowed cosine creates two sinc-shaped copies



A cosine inside a rectangle has spectrum centered around $\pm\omega_0$, not around zero.

Fourier transform of a spike

$$x(t) = \delta(t)$$



What does this single spike look like in the frequency domain?

Simple, let's use the sifting property!

$$\text{For } x(t) = \delta(t),$$

$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt.$$

Now the impulse simply samples the background function at $t = 0$:

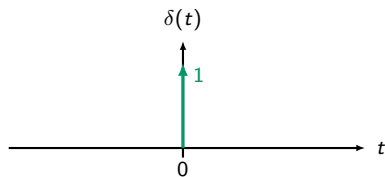
$$\int_{-\infty}^{\infty} \delta(t) f(t) dt = f(0).$$

$$X(j\omega) = e^{-j\omega \cdot 0} = 1.$$

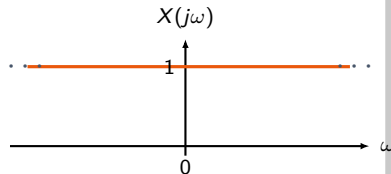
So the spectrum is completely flat

$$\delta(t) \longleftrightarrow 1$$

Time domain



Frequency domain



A very narrow spike in time needs all frequencies equally.

Small bonus: what if the spike is shifted?

$$\text{Let, } x(t) = \delta(t - t_0).$$

$$\text{Then, } X(j\omega) = \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j\omega t} dt.$$

Using sifting, the impulse samples the exponential at $t = t_0$:

$$X(j\omega) = e^{-j\omega t_0}.$$

$$\boxed{\delta(t - t_0) \longleftrightarrow e^{-j\omega t_0}}$$

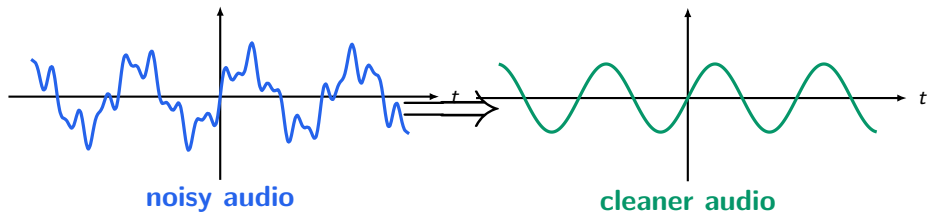
Applications

Where Fourier transform quietly appears

Audio Denoising

Remove the unwanted frequencies

Audio denoising: keep the voice, reduce the noise

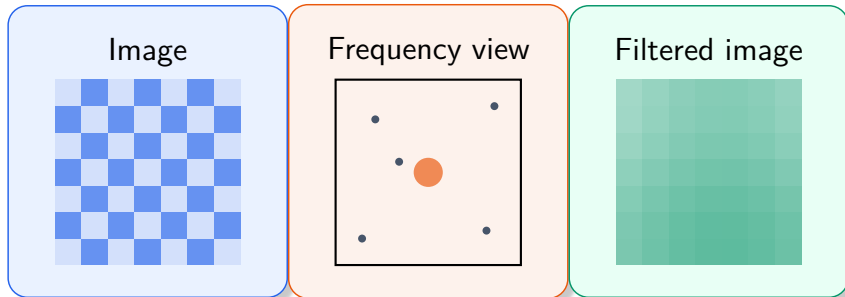


Fourier transform lets us see which frequencies are voice-like and which are noise-like.

Image Denoising

Clean an image by editing its frequencies

Image denoising: noise often hides in high frequencies

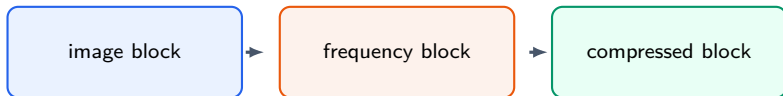


A blurry-but-clean image is often better than a sharp-but-noisy one.

JPEG Compression

Throw away what the eye barely notices

JPEG compression: not all frequencies are equally important



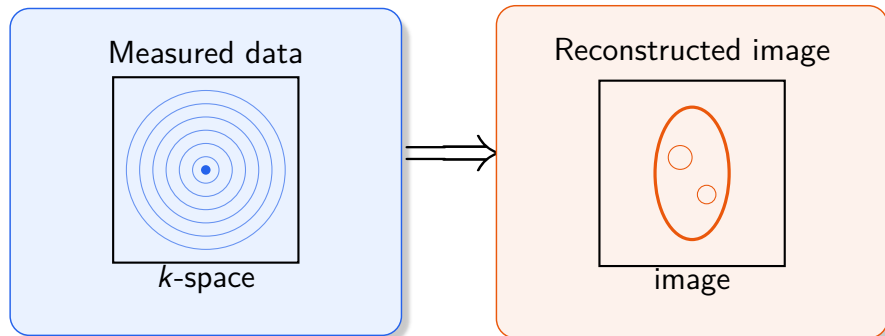
small high-frequency coefficients are rounded away

Fourier-like ideas help images become smaller without looking very different.

MRI

The scanner collects frequency-domain data first

MRI: the picture is reconstructed from frequency samples

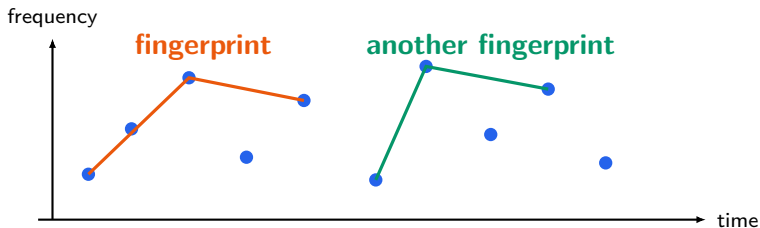


MRI is a beautiful real-life case where frequency domain comes before the image.

Music Recognition

Fingerprints in frequency

Music recognition: a song can be identified from frequency peaks



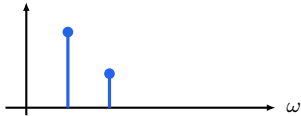
Even a short audio clip can reveal a unique pattern of strong frequencies.

Vibration Diagnosis

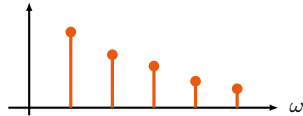
Machines complain in frequencies

Vibration diagnosis: a faulty machine changes its spectrum

Healthy



Faulty

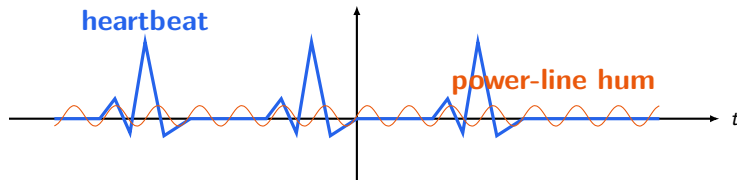


A bearing crack or loose part may be easier to see in frequency than in time.

ECG Filtering

Rescue the heartbeat from interference

ECG filtering: remove hum without destroying the heartbeat



Fourier transform helps design filters that remove unwanted frequencies carefully.

Thank you!

Thank youuu! Until we meet again...

